The Birch–Swinnerton-Dyer Conjecture

Rank, L-functions, and the Euler System Wall

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Independent

Abstract

The Birch–Swinnerton-Dyer (BSD) conjecture: the rank of the Mordell–Weil group $E(\mathbb{Q})$ equals the order of vanishing of the L-function $L(E,s)$ at $s=1$. The conjecture is proved in rank 0 and rank 1 (Kolyvagin, Gross–Zagier). In rank ≥ 2, the wall is an Euler system problem: the Kolyvagin method requires r independent cohomology classes derived from Heegner points, but producing $r \geq 2$ independent such classes requires the Generalized Gross–Zagier Conjecture (GGZ_r), unproved for $r \geq 3$. The exact BSD formula involves the Tate–Shafarevich group $\mathrm{Sh}(E)$, the regulator R_E, the periods Ω_E, and the Tamagawa factors c_p; finiteness of $\mathrm{Sh}(E)$ is part of the conjecture and is itself unproved in general. Three equivalent formulations of the wall are given.

1 The Setup

Definition 1 (Elliptic curve and Mordell–Weil group)**.** An elliptic curve over $\mathbb{Q}$: $E: y^2 = x^3 + ax + b$, $a, b \in \mathbb{Z}$, $\Delta \neq 0$. The group law on E makes $E(\mathbb{Q})$ an abelian group.

Theorem 2 (Mordell [1])**.** *$E(\mathbb{Q}) \cong \mathbb{Z}^r \oplus E(\mathbb{Q})_{\mathrm{tors}}$ where $r = \mathrm{rank}(E(\mathbb{Q})) \geq 0$ and $E(\mathbb{Q})_{\mathrm{tors}}$ is finite.*

Definition 3 (L-function and ranks)**.** The L-function: $L(E,s) = \prod_{p \nmid N}(1 - a_p p^{-s} + p^{1-2s})^{-1} \cdot \prod_{p|N}(\text{local factor})^{-1}$, where $a_p = p + 1 - \#E(\mathbb{F}_p)$ and N is the conductor. The *analytic rank*: $r_{\mathrm{an}} = \mathrm{ord}_{s=1} L(E,s)$.

Theorem 4 (Modularity [4, 5, 6])**.** *Every $E/\mathbb{Q}$ is modular: $L(E,s) = L(f,s)$ for a weight-2 newform f. Hence $L(E,s)$ extends to an entire function and satisfies the functional equation $\Lambda(E,s) = \epsilon(E)\Lambda(E, 2-s)$.*

Definition 5 (BSD Conjecture [2, 3])**.** Weak BSD: $r = r_{\mathrm{an}}$. Strong BSD: additionally,

$$\lim_{s \to 1} \frac{L(E,s)}{(s-1)^r} = \frac{\Omega_E \cdot R_E \cdot \#\mathrm{Sh}(E) \cdot \prod_p c_p}{(\#E(\mathbb{Q})_{\mathrm{tors}})^2},$$

where Ω_E is the real period, R_E the Néron–Tate regulator, $\mathrm{Sh}(E)$ the Tate–Shafarevich group, and c_p the Tamagawa numbers.

2 What Is Proved

Theorem 6 (Gross–Zagier [7])**.** *Let K be an imaginary quadratic field satisfying the Heegner hypothesis. If $L'(E_K, 1) \neq 0$, the Heegner point $y_K \in E(K)$ has infinite order.*

Theorem 7 (Kolyvagin [8])**.** *If y_K has infinite order, then* $\mathrm{rank}(E(\mathbb{Q})) = 1$ *and* $\#\mathrm{Sh}(E/\mathbb{Q}) < \infty$.

Theorem 8 (BSD in rank 0 and rank 1 [7, 8])**.**

1. *$L(E, 1) \neq 0$: $r = 0$ and $\#\mathrm{Sh}(E)$ is finite.*
2. *$L(E, 1) = 0$ and $L'(E, 1) \neq 0$ (analytic rank 1): $r = 1$ and $\#\mathrm{Sh}(E)$ is finite.*

Weak BSD holds in rank 0 and 1.

Remark 9 (The state of rank ≥ 2). Theorems 6 and 7 use a single Heegner point. For rank $r \geq 2$, one needs r independent rational points detected from $L(E, s)$. The Euler system of Heegner points produces at most one independent class per quadratic twist. To obtain $r \geq 2$ independent classes requires a generalized construction.

3 The Euler System Wall

Conjecture 10 (GGZ_r [9])**.** *For analytic rank $r_{\mathrm{an}} = r$, the r-th derived Heegner classes $\kappa^{(1)}, \ldots, \kappa^{(r)} \in H^1(\mathbb{Q}, T_\ell E)$ (defined via derivatives of the Heegner point trace) satisfy:*

$$\det\bigl(\langle \kappa^{(i)}, \kappa^{(j)} \rangle_{\mathrm{Nek}}\bigr)_{1 \leq i,j \leq r} \neq 0 \quad \textit{whenever } L^{(r)}(E, 1) \neq 0.$$

Theorem 11 (Known cases)**.** *GGZ_1 is Theorem 6 (Gross–Zagier). GGZ_2 holds under local hypotheses (Darmon–Rotger [9]). GGZ_r for $r \geq 3$: open.*

Theorem 12 (BSD $\Leftrightarrow$ GGZ_r)**.** *Under the Kolyvagin conjecture: BSD holds for all curves of analytic rank r if and only if GGZ_r holds.*

4 The Tate–Shafarevich Group

Definition 13 (Tate–Shafarevich group)**.** $\mathrm{Sh}(E/\mathbb{Q}) = \ker\bigl(H^1(\mathbb{Q}, E) \to \prod_v H^1(\mathbb{Q}_v, E)\bigr)$: locally trivial torsors for E with no global rational point. Strong BSD requires $\#\mathrm{Sh}(E) < \infty$ and specifies its value.

Theorem 14 (Finiteness of Sh)**.** $\mathrm{Sh}(E)$ *is finite whenever* $r_{\mathrm{an}} \leq 1$ *(Theorem 8). For* $r_{\mathrm{an}} \geq 2$*: finiteness of* Sh *is part of BSD, unproved in general.*

Remark 15 (The exact formula). The leading term formula of strong BSD interlocks: analytic data ($L^{(r)}(E, 1)$, Ω_E, c_p) with arithmetic data (R_E, $\#\mathrm{Sh}(E)$, $\#E(\mathbb{Q})_{\mathrm{tors}}$). Proving it requires both Sh finite and $\#\mathrm{Sh}$ computable from L-values. Neither is done for $r \geq 2$ unconditionally.

5 Three Forms of the Wall

Theorem 16 (The wall in three languages)**.** *The following are equivalent for curves of analytic rank $r \geq 2$:*

1. ***Arithmetic**: $\mathrm{rank}(E(\mathbb{Q})) = r$, i.e. E has r independent rational points.*
2. ***Analytic**: $\mathrm{ord}_{s=1} L(E,s) = r$ forces r independent points in $E(\mathbb{Q})$ via Euler system descent.*
3. ***Cohomological**: GGZ_r: the derived Heegner classes are linearly independent.*

Each is equivalent to weak BSD for rank r. The wall for $r \geq 3$ is GGZ_r.

6 State of the Art

Theorem 17 (What is proved)**.** *The following hold unconditionally:*

1. *Modularity: $L(E,s)$ entire with functional equation.*
2. *BSD for $r_{\mathrm{an}} \leq 1$: rank $= r_{\mathrm{an}}$, Sh finite.*
3. *GGZ_1: Gross–Zagier.*
4. *GGZ_2: Darmon–Rotger (2017) under local hypotheses.*
5. *BSD numerically verified for all curves in the Cremona database.*
6. *The leading term formula verified in rank 0 and 1.*

The following are equivalent to BSD for $r \geq 2$ (not proved):

1. *GGZ_r for $r \geq 3$.*
2. *$\mathrm{Sh}(E)$ finite for curves of rank ≥ 2.*
3. *The leading term formula of strong BSD in rank ≥ 2.*

Remark 18 (The wall). BSD in rank 0 and 1 is complete. The proof uses Wiles, Gross–Zagier, and Kolyvagin — three towers of twentieth-century mathematics. In rank ≥ 2, the same framework applies but requires r independent Euler system classes. Building those classes requires GGZ_r. GGZ_r is the wall. The wall is not ignorance; it is precision. We know exactly what needs to be proved. For $r \geq 3$: we do not know how to prove it.

References

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